EE515: FUNDAMENTALS OF VLSI ARCHITECTURE ASSIGNMENT

IMPLEMENTING A DA-BASED 16 POINT DFT IN VERILOG

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EEE - VLSI & NANOELECTRONICS

Contents

1 Introduction

The Discrete Fourier Transform (DFT) converts a sequence of time-domain samples into frequency-domain components. The mathematical expression for an N-point DFT is given by

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \tag{1}$$

where

$$W_N = e^{-j\frac{2\pi}{N}} \tag{2}$$

For a 16-point DFT,

$$X(k) = \sum_{n=0}^{15} x(n) W_{16}^{kn} \tag{3}$$

where

$$W_{16} = e^{-j\frac{2\pi}{16}} \tag{4}$$

1.1 16-Point DFT Matrix Representation

The 16-point DFT can be represented in matrix form as

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ \vdots \\ X(15) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & W_{16}^{1} & W_{16}^{2} & \cdots & W_{16}^{15} \\ 1 & W_{16}^{2} & W_{16}^{4} & \cdots & W_{16}^{30} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_{16}^{15} & W_{16}^{30} & \cdots & W_{16}^{225} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ \vdots \\ x(15) \end{bmatrix}$$

1.2 Distributed Arithmetic (DA) Based Formulation

Distributed Arithmetic (DA) eliminates explicit multipliers by using precomputed sums stored in ROM.

Each input sample is represented in binary form as

$$x(n) = \sum_{b=0}^{B-1} x_b(n) 2^{-b} \tag{5}$$

Substituting into the DFT equation,

$$X(k) = \sum_{n=0}^{15} \left(\sum_{b=0}^{7} x_b(n) 2^{-b} \right) W_{16}^{kn} \tag{6}$$

(here, x is of 8 bits) Rearranging,

$$X(k) = \sum_{b=0}^{7} 2^{-b} \left[\sum_{n=0}^{15} x_b(n) W_{16}^{kn} \right] \tag{7}$$

1.3 Twiddle Factor Values

The twiddle factor for 16-point DFT is

$$W_{16}^{k} = e^{-j\frac{2\pi k}{16}} \tag{8}$$

The twiddle factors are listed below:

Table 1: Twiddle Factor Table for 16-Point DFT

k	W_{16}^{k}	Numerical Value
0	W_{16}^{0}	$1 + j0$
1	W_{16}^{1}	$0.9239 - j0.3827$
2	W_{16}^{2}	$0.7071 - j0.7071$
3	W_{16}^{3}	$0.3827 - j0.9239$
4	W_{16}^{4}	$0 - j1$
5	W_{16}^{5}	$-0.3827 - j0.9239$
6	W_{16}^{6}	$-0.7071 - j0.7071$
7	W_{16}^{7}	$-0.9239 - j0.3827$
8	W_{16}^{8}	$-1 + j0$
9	W_{16}^{9}	$-0.9239 + j0.3827$
10	W_{16}^{10}	$-0.7071 + j0.7071$
11	W_{16}^{11}	$-0.3827 + j0.9239$
12	W_{16}^{12}	$0 + j1$
13	W_{16}^{13}	$0.3827 + j0.9239$
14	W_{16}^{14}	$0.7071 + j0.7071$
15	W_{16}^{15}	$0.9239 + j0.3827$

1.4 Twiddle Factor ROM Structure

Table 2: DA-Based ROM Look-Up Table

Address	Binary Address	ROM Output
0	0000	0
1	0001	twf_0
2	0010	twf_1
3	0011	$twf_1 + twf_0$
4	0100	twf_2
5	0101	$twf_2 + twf_0$
6	0110	$twf_2 + twf_1$
7	0111	$twf_2 + twf_1 + twf_0$
8	1000	twf_3
9	1001	$twf_3 + twf_0$
10	1010	$twf_3 + twf_1$
11	1011	$twf_3 + twf_1 + twf_0$
12	1100	$twf_3 + twf_2$
13	1101	$twf_3 + twf_2 + twf_0$
14	1110	$twf_3 + twf_2 + twf_1$
15	1111	$twf_3 + twf_2 + twf_1 + twf_0$

2 Circuit Diagram

The DA-based 16-point DFT architecture is implemented using ROM-based Distributed Arithmetic. The complete architecture consists of the following major blocks:

- Input Registers
- Shift Registers
- Twiddle Factor ROM
- Adder Network
- Shift Accumulator
- Output Registers

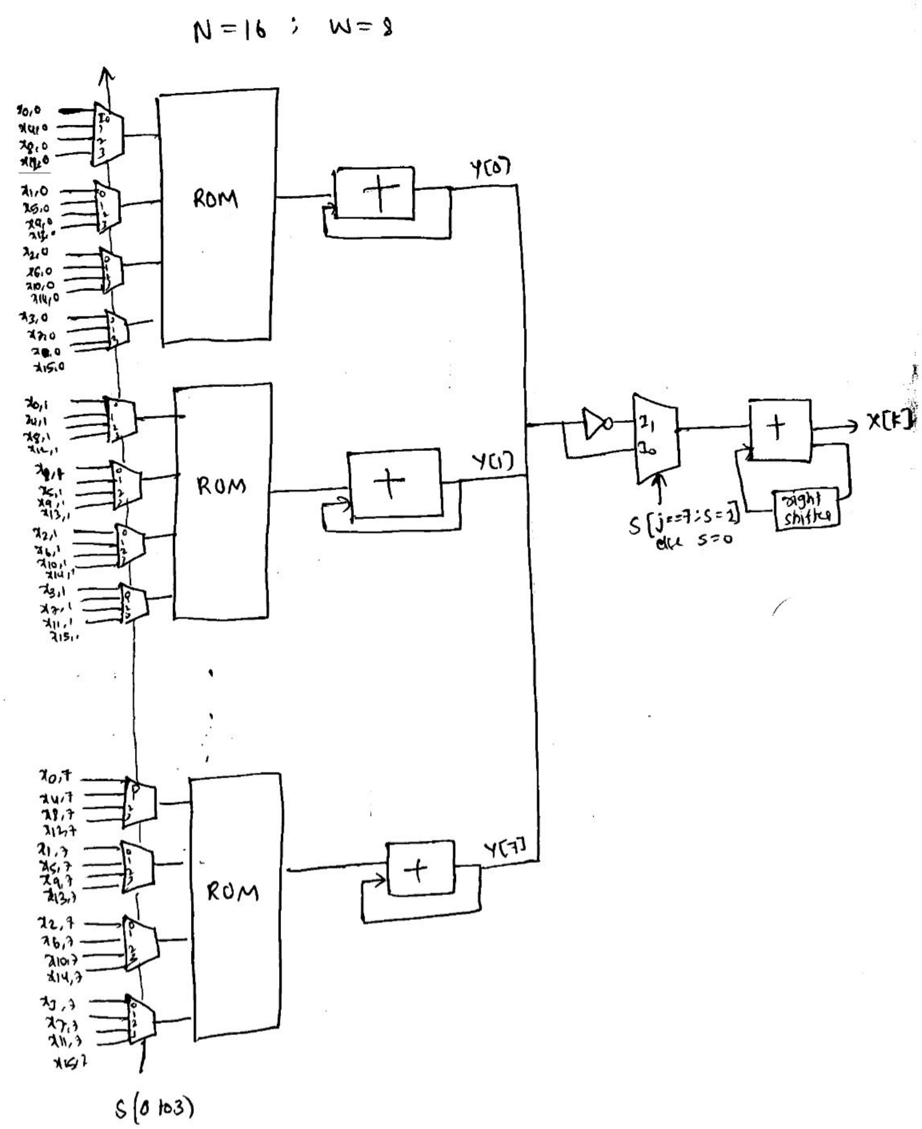


Figure 1: DFT Architecture which computes one DFT output X[k]

3 Simulation Results

3.1 Test Case 1 :

$$x = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$$

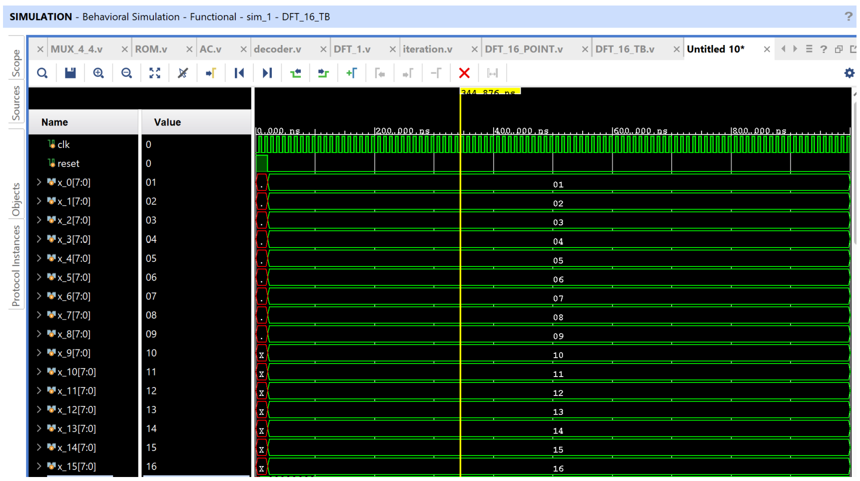


Figure 2: inputs of DFT

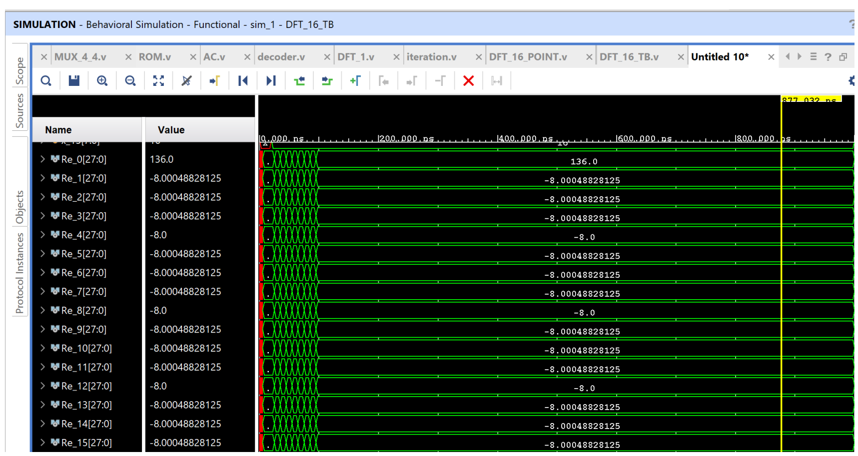


Figure 3: DFT output: real part

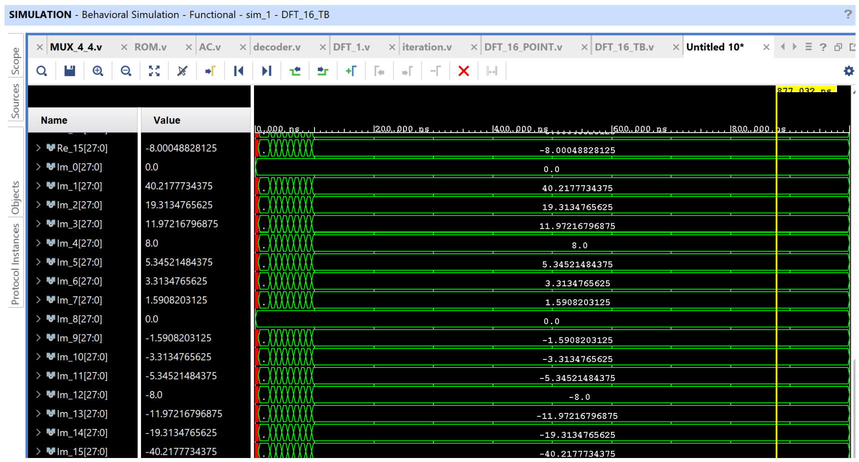
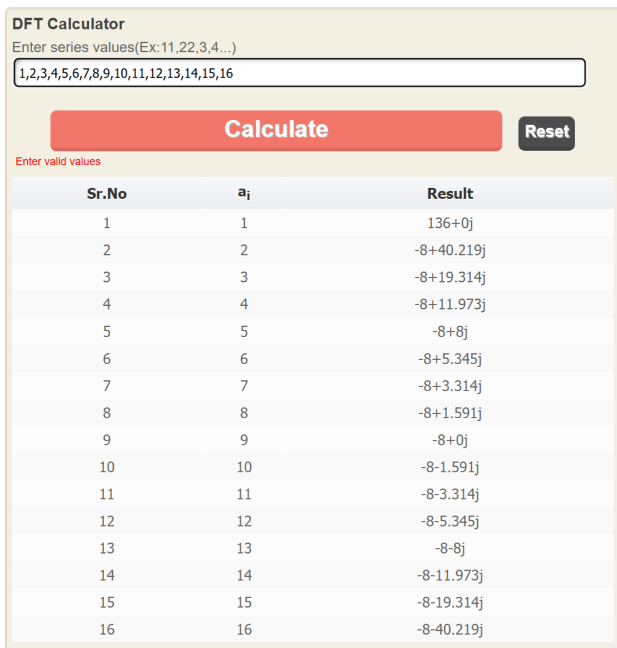


Figure 4: DFT output : imaginary part

```
# run 1000ns
X[0] = 136.000000 + i(0.000000)
X[1] = -8.000488 + i(40.217773)
X[2] = -8.000488 + i(19.313477)
X[3] = -8.000488 + i(11.972168)
X[4] = -8.000000 + i(8.000000)
X[5] = -8.000488 + i(5.345215)
X[6] = -8.000488 + i(3.313477)
X[7] = -8.000488 + i(1.590820)
X[8] = -8.000000 + i(0.000000)
X[9] = -8.000488 + i(-1.590820)
X[10] = -8.000488 + i(-3.313477)
X[11] = -8.000488 + i(-5.345215)
X[12] = -8.000000 + i(-8.000000)
X[13] = -8.000488 + i(-11.972168)
X[14] = -8.000488 + i(-19.313477)
X[15] = -8.000488 + i(-40.217773)
INFO: [USF-XSim-96] XSim completed. Design snapshot 'DFT_16_TB_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
```

Figure 5: output of DFT



Sr.No	a_i	Result
1	1	136+0j
2	2	-8+40.219j
3	3	-8+19.314j
4	4	-8+11.973j
5	5	-8+8j
6	6	-8+5.345j
7	7	-8+3.314j
8	8	-8+1.591j
9	9	-8+0j
10	10	-8-1.591j
11	11	-8-3.314j
12	12	-8-5.345j
13	13	-8-8j
14	14	-8-11.973j
15	15	-8-19.314j
16	16	-8-40.219j

Figure 6: Expected output of DFT

post synthesis simulation results

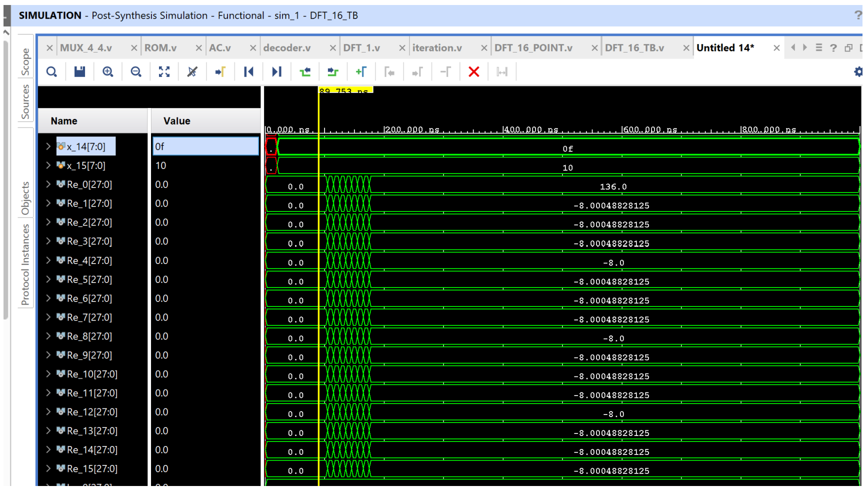


Figure 7: DFT output : real part

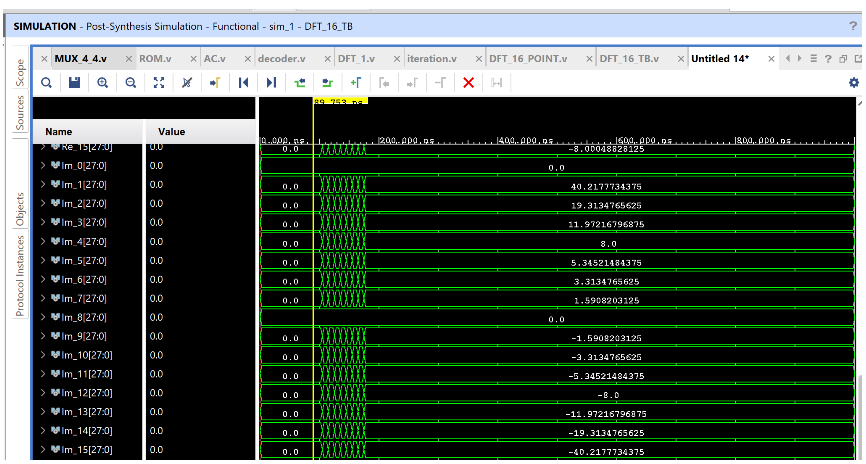


Figure 8: DFT output : imaginary part

3.2 Test Case 2 :

$$x = \{1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0\}$$

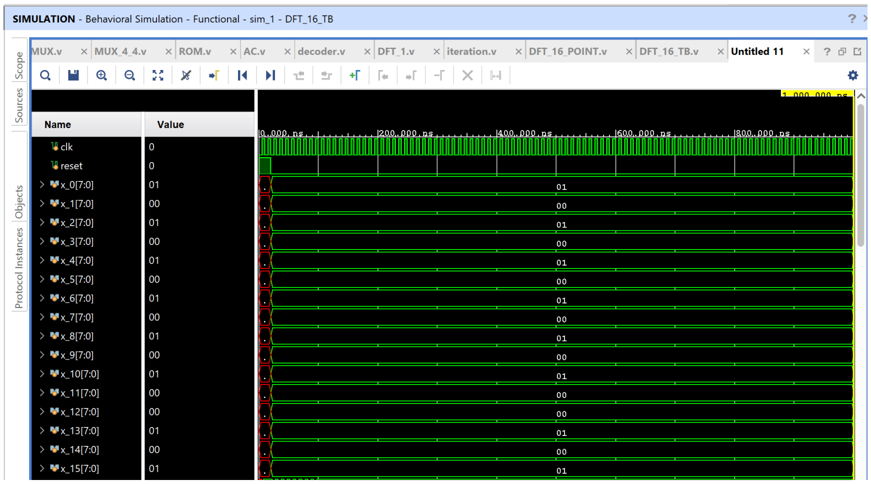


Figure 9: inputs of DFT

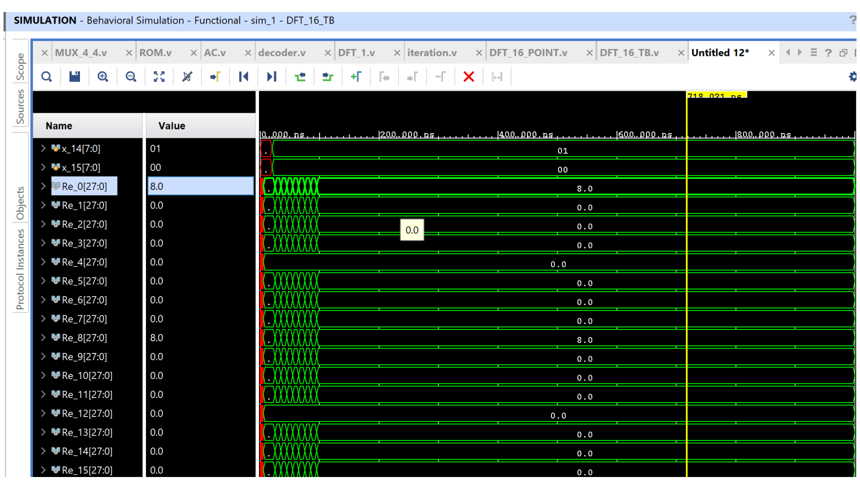


Figure 10: DFT output: real part

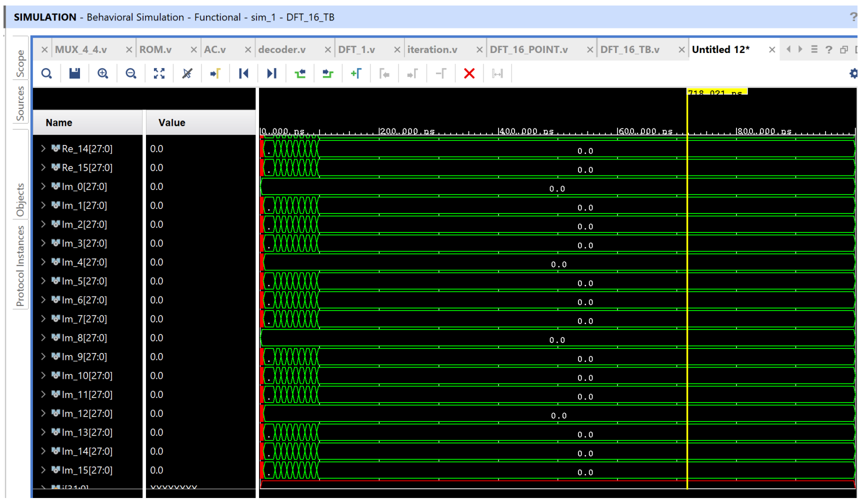
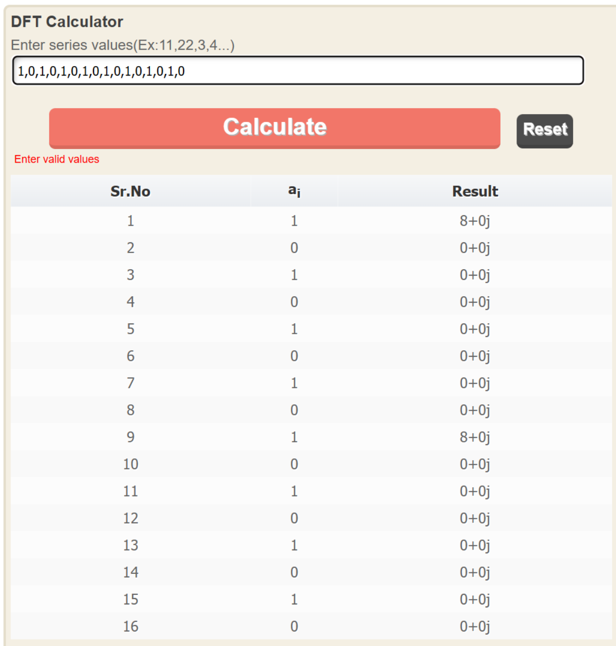


Figure 11: DFT output : imaginary part

```
# }
# run 1000ns
X[0] = 8.000000 + i(0.000000)
X[1] = 0.000000 + i(0.000000)
X[2] = 0.000000 + i(0.000000)
X[3] = 0.000000 + i(0.000000)
X[4] = 0.000000 + i(0.000000)
X[5] = 0.000000 + i(0.000000)
X[6] = 0.000000 + i(0.000000)
X[7] = 0.000000 + i(0.000000)
X[8] = 8.000000 + i(0.000000)
X[9] = 0.000000 + i(0.000000)
X[10] = 0.000000 + i(0.000000)
X[11] = 0.000000 + i(0.000000)
X[12] = 0.000000 + i(0.000000)
X[13] = 0.000000 + i(0.000000)
X[14] = 0.000000 + i(0.000000)
X[15] = 0.000000 + i(0.000000)
INFO: [USF-XSim-96] XSim completed. Design snapshot 'DFT_16_TB_behav' loaded.
```

Figure 12: output of DFT



Sr.No	a_i	Result
1	1	8+0j
2	0	0+0j
3	1	0+0j
4	0	0+0j
5	1	0+0j
6	0	0+0j
7	1	0+0j
8	0	0+0j
9	1	8+0j
10	0	0+0j
11	1	0+0j
12	0	0+0j
13	1	0+0j
14	0	0+0j
15	1	0+0j
16	0	0+0j

Figure 13: Expected output of DFT

post synthesis simulation results

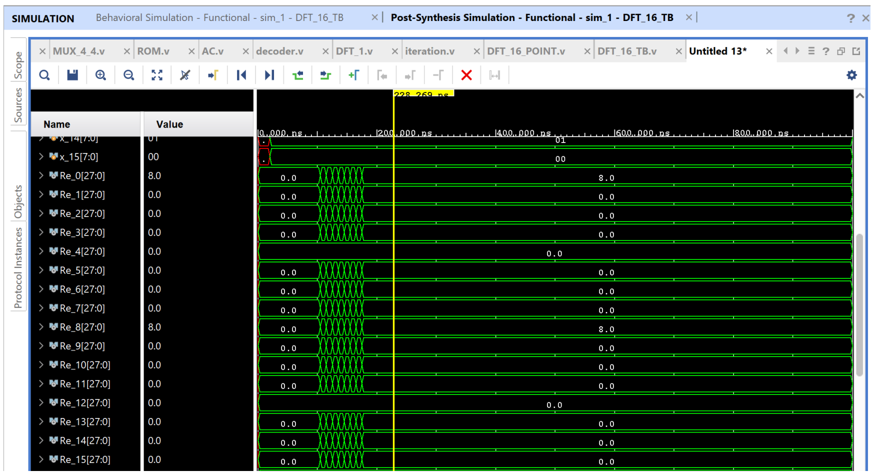


Figure 14: DFT output : real part

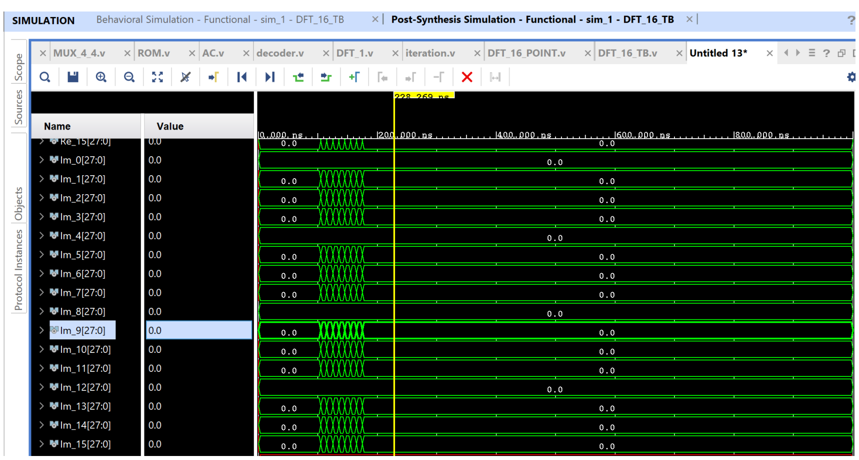


Figure 15: DFT output : imaginary part

3.3 Test Case 3 :

$$x = \{8, 7, 6, 3, 0, -3, -6, -7, -8, -7, -6, -3, 0, 3, 6, 7\}$$

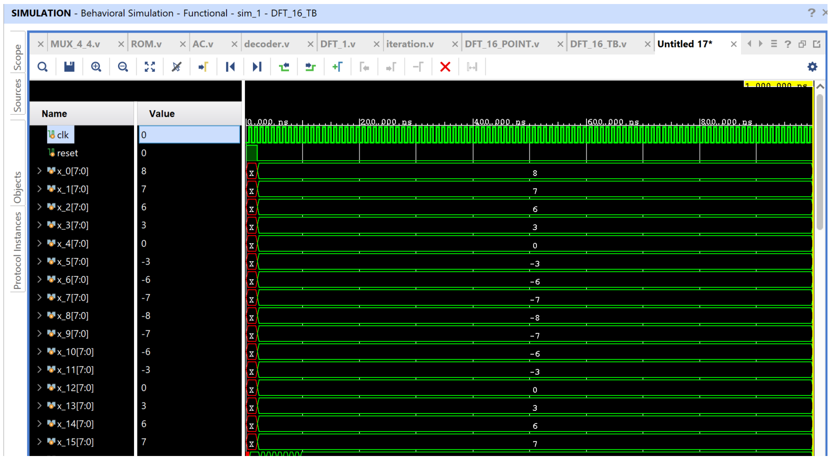


Figure 16: inputs of DFT

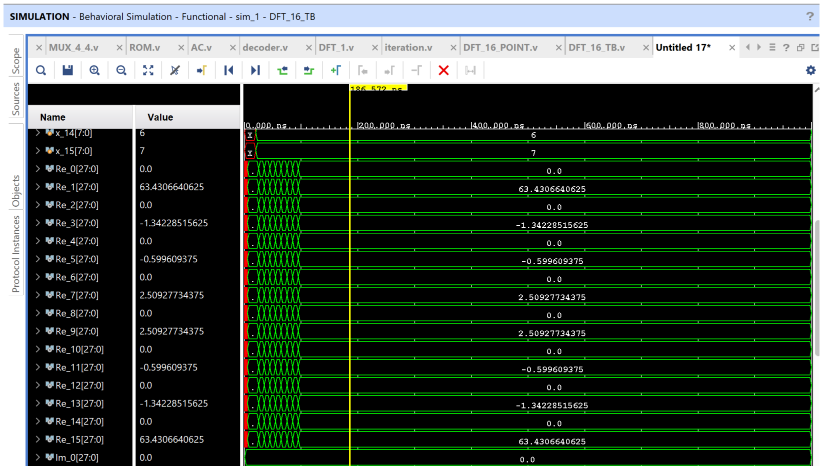


Figure 17: DFT output: real part

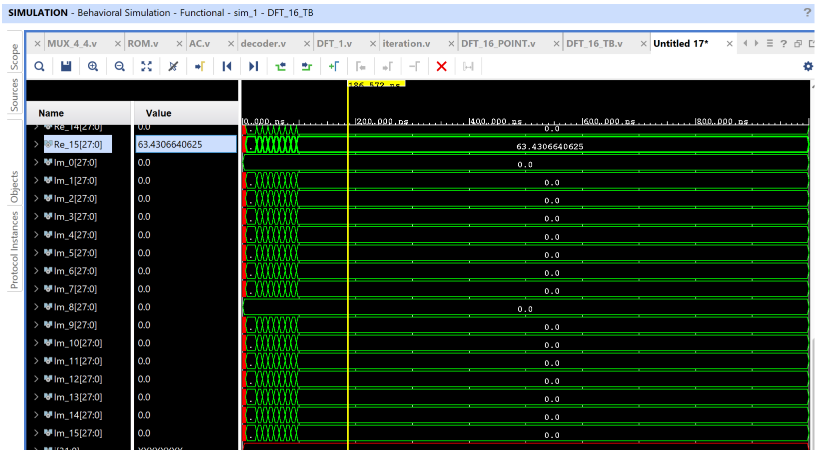


Figure 18: DFT output : imaginary part

```
# run 1000ns
X[0] = 0.000000 + i(0.000000)
X[1] = 63.430664 + i(0.000000)
X[2] = 0.000000 + i(0.000000)
X[3] = -1.342285 + i(0.000000)
X[4] = 0.000000 + i(0.000000)
X[5] = -0.599609 + i(0.000000)
X[6] = 0.000000 + i(0.000000)
X[7] = 2.509277 + i(0.000000)
X[8] = 0.000000 + i(0.000000)
X[9] = 2.509277 + i(0.000000)
X[10] = 0.000000 + i(0.000000)
X[11] = -0.599609 + i(0.000000)
X[12] = 0.000000 + i(0.000000)
X[13] = -1.342285 + i(0.000000)
X[14] = 0.000000 + i(0.000000)
X[15] = 63.430664 + i(0.000000)
INFO: [USF-XSim-96] XSim completed. Design snapshot 'DFT_16_TB_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
```

Figure 19: output of DFT

Sr.No	a_i	Result
1	8	0+0j
2	7	63.431+0j
3	6	0+0j
4	3	-1.342+0j
5	0	0+0j
6	-3	-0.599+0j
7	-6	0+0j
8	-7	2.51+0j
9	-8	0+0j
10	-7	2.51+0j
11	-6	0+0j
12	-3	-0.599+0j
13	0	0+0j
14	3	-1.342+0j
15	6	0+0j
16	7	63.431+0j

Figure 20: Expected output of DFT

post synthesis simulation results

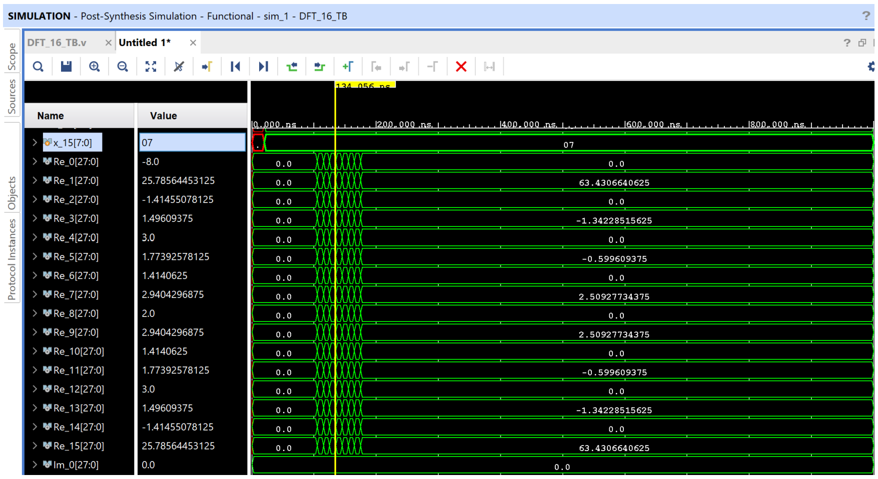


Figure 21: DFT output : real part

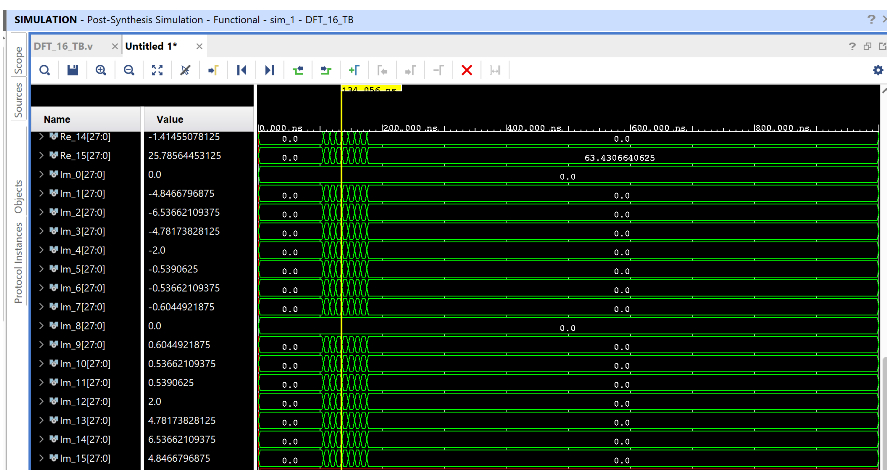


Figure 22: DFT output : imaginary part

4 Power

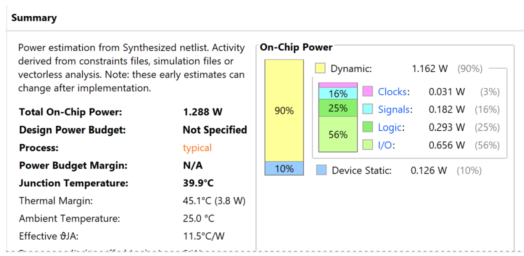


Figure 23: Power consumption

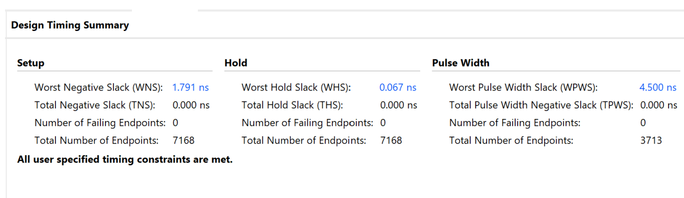


Figure 24: Timing summary